higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 4

(12020514)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Reinforcing bars that are delivered at building site should be stored in some logical manner.

Explain how you will store the bundles of reinforcing bars that were delivered on a building site, in a logical manner to maintain efficient stock control. (4)

1.2 Complete the following sentences by filling the missing word(s). Write only the word(s) next to the question number (1.2.1–1.2.6) in the ANSWER BOOK.

water; neutralise; alkaline; oxide; corrode; active; clay; acidity; limestone; passive; chlorides

Alkali is one of a class of substance that (1.2.1) …. and is opposite of (1.2.2)….. Portland cement consists of a mixture of (1.2.3)….. and (1.2.4)….. The chemical environment provided by cement is (1.2.5) …. and causes a (1.2.6) … ferric oxide protective film on the surface of the steel.

(6 × 1) (6)
[10]

QUESTION 2

2.1 Explain why we must always cut off ends of the wire when fixing steel. (3)

2.2 Name TWO personal protective equipment that one has to wear when cutting and bending steel. (2)

2.3 When installing reinforcing steel, different types of ties are used to tie the rebars together.

Illustrate by means of neat sketches the following ties.

2.3.1 Crown tie

2.3.2 Slash tie

2.3.3 Ring hairpin tie

2.3.4 Splice tie

2.3.5 Hairpin tie

(5 × 2) (10)

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2.4 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'true' or 'false' next to the question number (2.4.1 - 2.4.5) in the ANSWER BOOK. Correct the statement if it is FALSE.

2.4.1 Hardened steel wire is used by most contractors because it is malleable and can easily be cut by hand

2.4.2 Ring slash ties are used to tie the vertical and horizontal bars together to prevent the vertical bars slipping down

2.4.3 Crown ties are strong and secure ties and are often used in the same way as ring hairpin ties.

2.4.4 When tying a slash tie you must wrap the wire diagonally once around the two crossing bars with the two ends on top

2.4.5 Epoxy-coated reinforcing bars must be fastened with zinc-coated tie wire in order to protect the epoxy-coating from being damaged.

(8)

2.5 Explain the use of the following tools that are used by a steel fixer in the construction industry:

2.5.1 Tying wire nip (2)

2.5.2 Bar tie twister (2)

2.6 Briefly explain FOUR factors that must be kept in mind when tying reinforcement. (4 × 2) (8)

[35]

QUESTION 3

3.1 Name TWO loads and pressures that falsework must withstand. (2)

3.2 Describe the sequence of description for reinforcement as it appears on drawings.

3.2.1 20Y10 (2)

3.2.2 4R8 (2)

3.3 According to the SANS 10100-2 one of the checks that must be performed before concreting can take place is to check that all the reinforcement bars are clean and contamination free.

Discuss the statement in relation to the SANS 10100-2 Act. (4)

[10]

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QUESTION 4

4.1 Draw and name THREE different types of slumps that can be achieved when determining the workability of concrete. (10)

4.2 The table below indicates the different mixing times of a Drum type mixer and a Pan type mixer. Complete the table below by filling in the missing information. Write only the information next to the question number (4.2.1–4.2.3) in the ANSWER BOOK.

	DRUM TYPE MIXER	**PAN TYPE MIXER**
Batch size	1.5 m³ or less	
Minimum mixing time	4.2.1……	1 minute
	Increase by 20 s for each additional cubic meter or part thereof.	4.2.2……..
Maximum continuous mixing time	4.2.3……	6 minutes

(3 × 1) (3)

4.3 Name and describe THREE different methods you can use to delay the stiffening of concrete. (12)
[25]

QUESTION 5

5.1 Give SIX advantages of preventing damage rather than trying to repair damaged concrete surfaces when making good. (6)

5.2 Explain the term *surface plucking*. (2)

5.3 Define the term *dry-pack mortar* (2)
[10]

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QUESTION 6

6.1 Give THREE general guidelines to follow when storing timber panels. (3)

6.2 Give TWO reasons how damages may occur to formwork when not in use. (2)

6.3 Name FIVE facts to be considered when storing small components in a weatherproof shed or containers. (5)

[10]

TOTAL: 100

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